SHADOWDESK

Treasury execution for Canton node operators

Limits the agent cannot break, enforced by the ledger.

CANTON NETWORK · OCTOBER 2026 · GITHUB.COM/SABER1Y/SHADOWDESK

THE PROBLEM, AS WE UNDERSTAND IT

Node operators earn Canton Coin but pay their bills in dollars

Regular conversions

Validator and app rewards arrive in CC. Hosting, staff and services are paid in dollars, so part of the income is converted again and again.

One key, no hard limit

Whoever runs the conversion, a person or a bot, holds the keys. Nothing stops them selling more, cheaper, or to someone nobody approved.

The limited party enforces the limit

When the rules live in the seller’s own code or runbook, a bug or a compromised key bypasses them.

A working hypothesis, not yet validated: we are testing it in conversations with operators. · 2

THE MANDATE

Two approvers sign the limits on the ledger

The treasury owner and a second approver sign a TreasuryMandate. The contract refuses a mandate where both are the same party.

Every request the agent opens must be co-signed by that second approver, which only the mandate can do.

WHAT A MANDATE FIXES	
Approved desks	dealer A, dealer B
Asset sold	one instrument
Asset received	one instrument
Size per request	maximum
Worst price	maximum
Expiry	date and time

The cap applies to each request; a cap on total sales across requests is not built yet.

ONE CONVERSION

Approve, quote, award, settle

01

Open

The agent opens a request from the mandate. Anything over its limits is refused.

02

Quote

Each approved desk quotes. A desk never sees a rival’s quote.

03

Award

The best price wins by a fixed rule. A request is awarded at most once.

04

Settle

The mandate is checked again, then both sides move in one transaction, or neither does.

THE GUARANTEE

The agent cannot open a request outside the mandate.

Our first version let the buyer create a request without a mandate and settle 5,000,000 at 149 against a mandate of 1,000,000 at 101. We found it, fixed it in the contracts, and now test the bypass four ways.

WHY CANTON

The limit is checked where the money moves

Private quotes

Visibility follows each contract’s parties, so a rival desk’s node never receives another desk’s price.

One transaction

The mandate check, the receipt and both Token Standard transfers commit together, or none of them do.

Already there

Node operators already run Canton parties and hold Canton Coin, so there is no new chain or wallet to adopt.

WHERE EACH CLAIM IS PROVEN

Two environments, each proving different things

Environment	Setup	What it proves
Canton DevNet	One participant, fauceted CBTC and BETH test tokens	Two rounds on the mandated package, 2026-10-25: each desk won once, each settled in one update
LocalNet	Two participants, synthetic assets	A losing desk on a separate node sees 0 of the winner's quotes
Daml suite	11 scenarios	Mandate limits, the four bypass attempts, at-most-once award, atomic settlement

No single run proves cross-node privacy and real-token settlement together.

Update ids: docs/evidence/devnet-2026-10-25-mandated.json · 7

REAL TODAY, AND NOT YET

The engine runs; the operators' trade does not yet

Piece	Status
Mandate enforced on every request and at settlement	Runs on DevNet
Sealed quotes, fixed award rule, one-transaction settlement	Runs on DevNet, test tokens
Selling Canton Coin for a dollar stablecoin	Not yet: CC and USDCx legs
A cap on total sales and a monthly budget	Not yet: cap is per request
Real desks quoting, real users	None yet

WHAT WE ARE TESTING

Five operator conversations before more building

0
conversations so far

5
target

WE NEED TO LEARN

1. Do operators convert Canton Coin to dollars regularly, and how much?
2. Who holds the keys to do it, and what limits them?
3. Has a conversion ever gone wrong?
4. Would a second approver and a ledger-enforced limit change how much they automate?
5. Which desks would quote, and do they want quotes private?

NEXT

If operators confirm the problem

01

Sell real Canton Coin through a CC allocation leg

02

Receive a dollar token, USDCx when it reaches DevNet

03

A total cap and a monthly budget on each mandate

04

Scheduled conversions within that budget

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